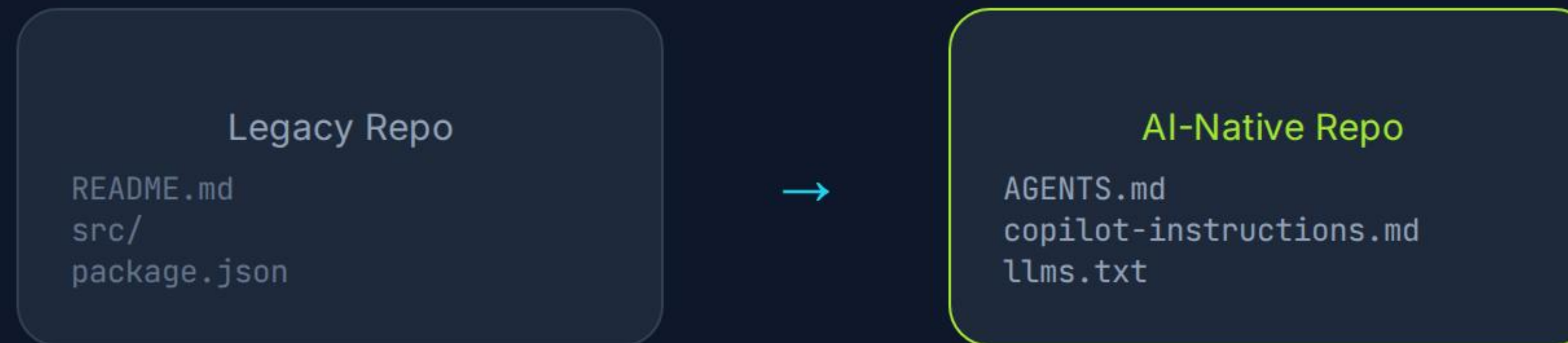


Context Engineering

Building **AI-Native** Repositories



"70% of AI coding tasks fail because the context is wrong. Not the model."

Parametric vs. Grounded Knowledge



Parametric Knowledge

Derived from training data. High-level syntax, patterns, and general libraries. Fixed at training cutoff. The "Stale" foundation.



Grounded Context

The "Truth" in your repository. Private APIs, naming conventions, and local state. **Real-time reconciliation.**

Decision Gap: Vercel found agents skip framework docs **44% of the time** because they think they already know the answer.

The Silent Killer: Context Rot

Context Window (1M Tokens) Decay

Safe Zone (0-50%): Instructions Retained

Compression (50-80%): Silent Summarization

Danger (80-95%): Conventions Discarded

Context Rot (95%+): Conventions Lost entirely

The Ticking Bomb

Even with large windows, agents auto-compact. **Architecture rules are the first to go.**

- Silent failure mode
- Code compiles but violates patterns
- Tests pass; Architecture fails

Push vs. Pull: Context Delivery

Passive Steering (Push)

- AGENTS.md
- copilot-instructions.md
- .instructions.md

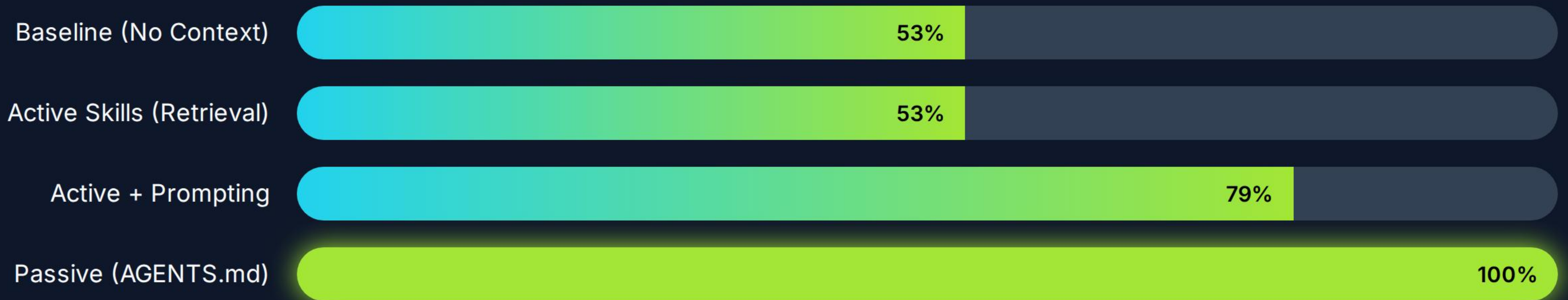
Always Present in System Prompt. Eliminates the Decision Gap. High reliability for small context.

Active Retrieval (Pull)

- read_file
- grep_search
- semantic_search

Agent Decides what to search. Necessary for scale. Vulnerable to "skipping" documentation.

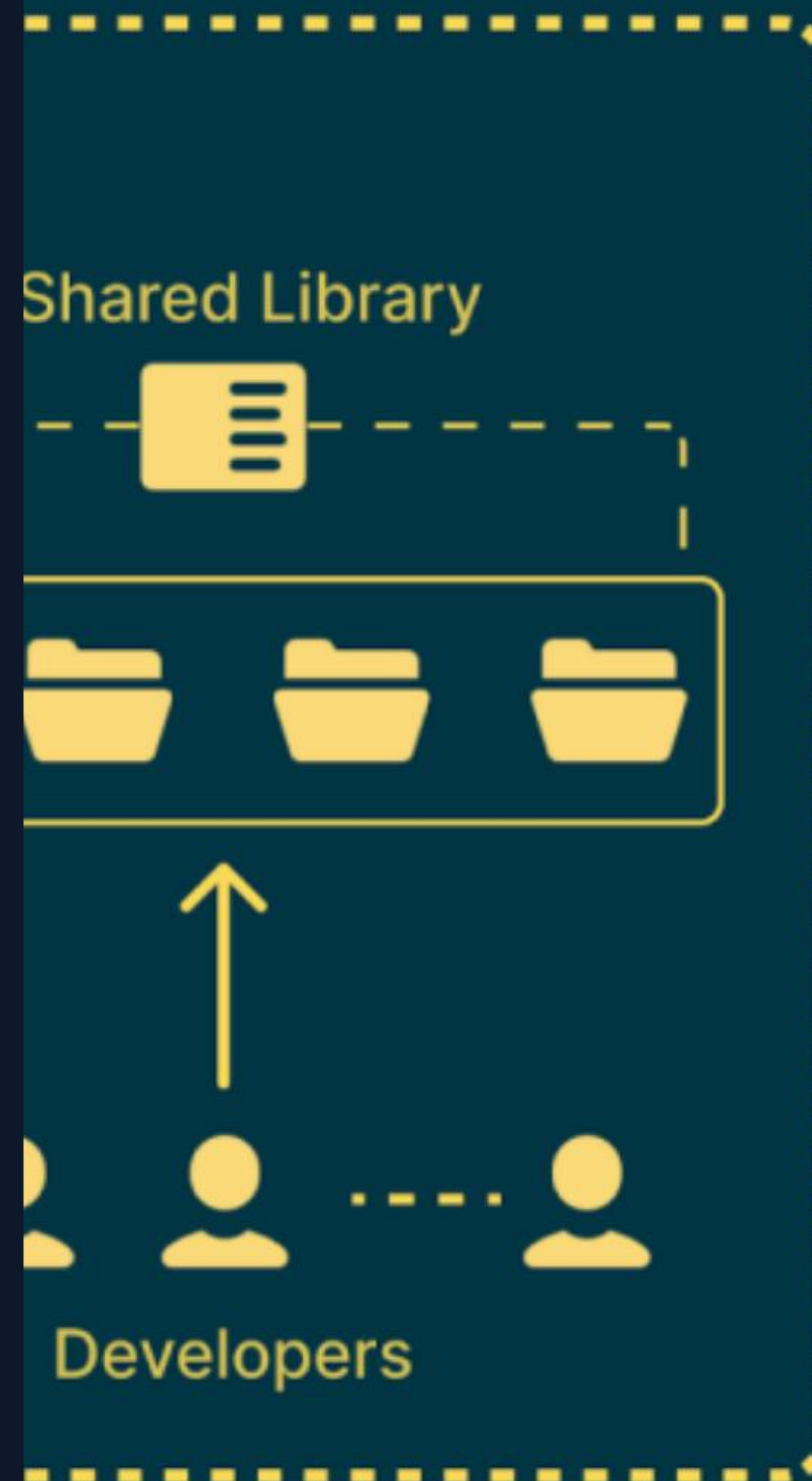
| The Decision Gap Impact



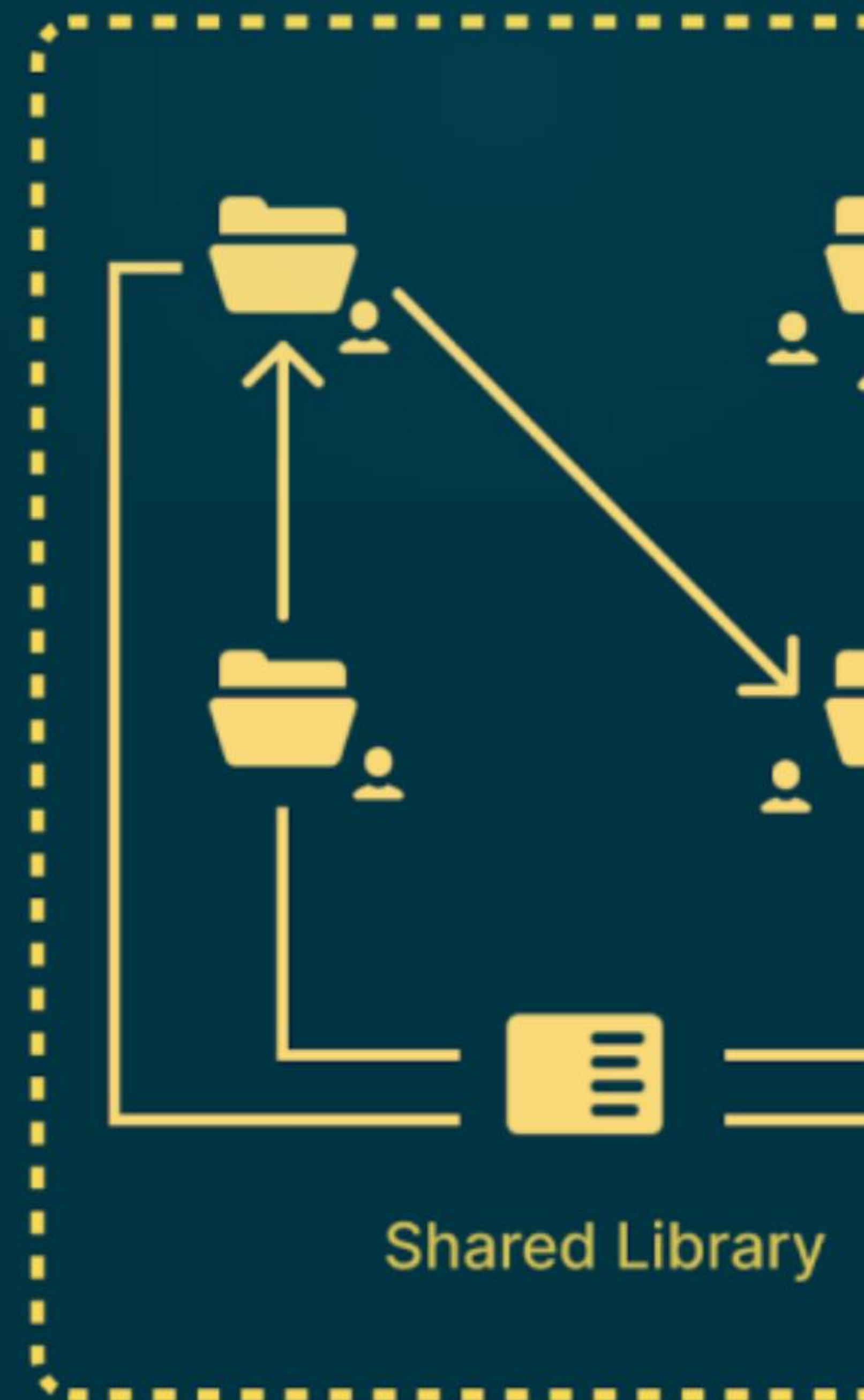
Passive context (AGENTS.md) removed the decision point entirely, reaching 100% success.

The Five Pillars

- 01 Unified Knowledge: Single source of truth in AGENTS.md
- 02 Scoped Precision: Glob-matched .instructions.md
- 03 Monorepo Mastery: Nested routing with precedence
- 04 AI-Readable Docs: llms.txt + structured README
- 05 Living Maintenance: Validation & audit loops



Monorepo



Polyrepo

Anthropic's **Progressive Disclosure**

LEVEL 1 **Frontmatter**

~100 tokens. **Always loaded.**
Match logic & Metadata.

LEVEL 2 **Skill Body**

Loaded **on match**. Step-by-step
instructions and examples.

LEVEL 3 **Resources**

Loaded **on demand**. Full API
guides and reference scripts.

Scale expertise without bloating the system prompt.

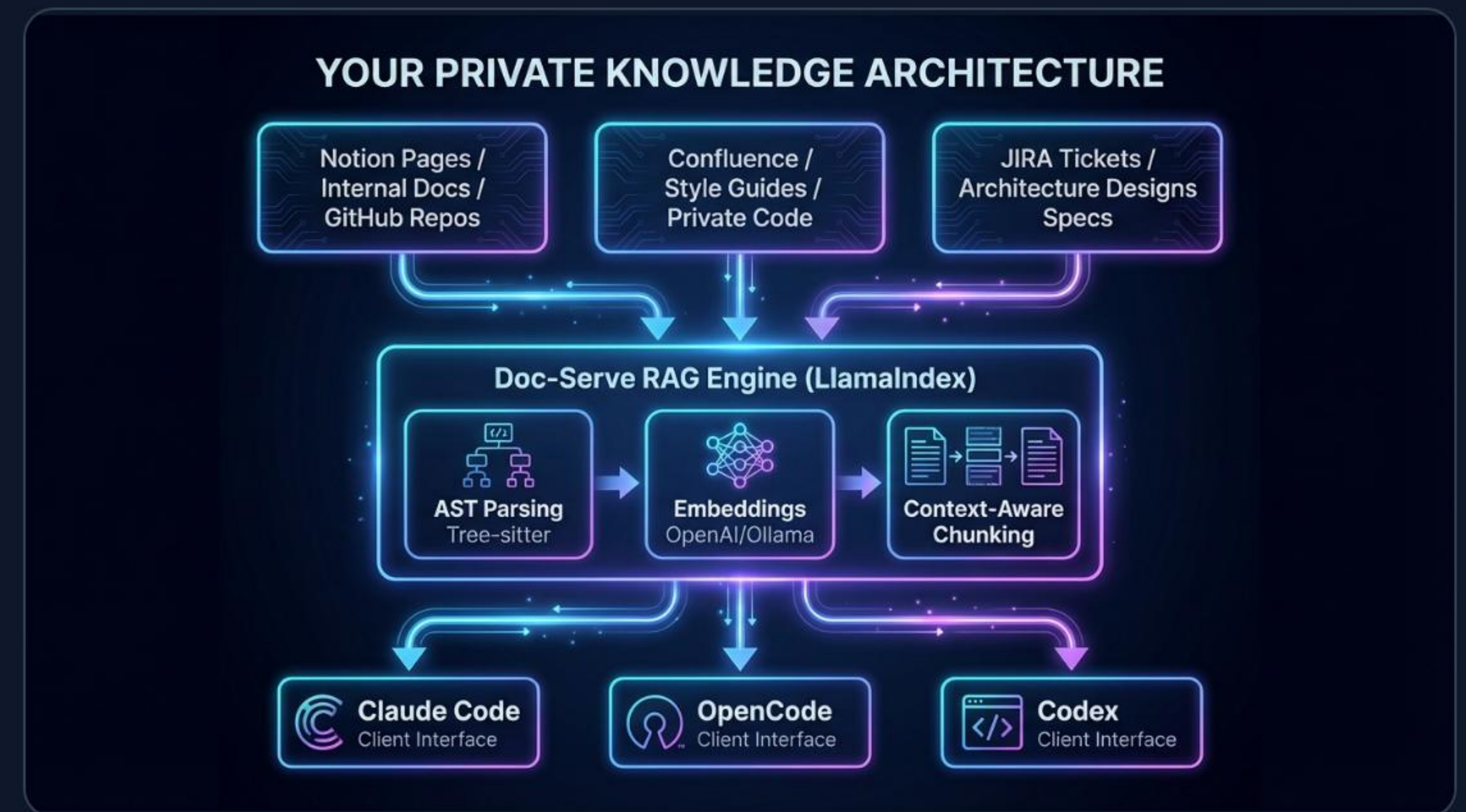
Architecture: MCP + Skills

The Kitchen Analogy

MCP (Model Context Protocol): The Kitchen. Provides tools, ingredients, and data access connections.

Skills: The Recipes. Defines *how* to use those tools for reliable outcomes.

"MCP gives access; Skills give deterministic execution."



Copilot vs. Devin

GitHub Copilot

The **Silent Partner**.

- IDE Plugin Architecture
- Sub-second latency
- Passive: copilot-instructions.md
- Surgical precision

Cognition Devin

The **Autonomous Engineer**.

- Cloud-based VM Agent
- Full Terminal/FS access
- Passive: AGENTS.md onboarding
- Large-scale refactors

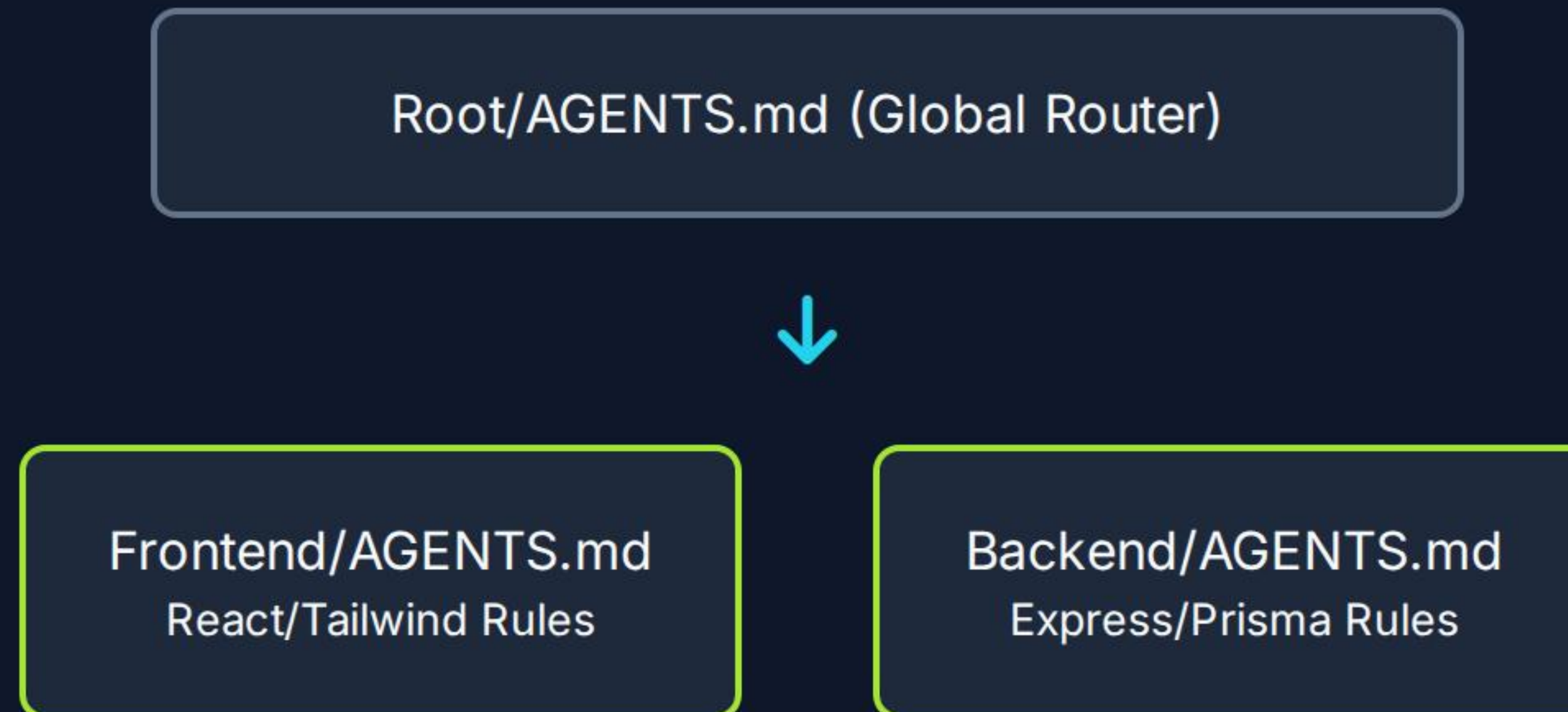
Complementary Tools: Use Copilot for flow, Devin for heavy lifting.

| Academic Reality Check

Metric	Vercel (Jan 2026)	ETH Zürich (arXiv 2602.11988)
Success Rate Change	+47% (53% → 100%)	Marginal effect
Knowledge Gap	Knowledge Absent from training	Knowledge Present in existing docs
Conclusion	Transformative for new APIs	Redundant for well-documented repos

Insight: Context files matter most for *Tribal Knowledge* and new version APIs.

Monorepo Precedence



Nearest-File Precedence: When editing packages/frontend, the agent loads frontend rules first, then root. Backend rules are never leaked.

The Recommended Strategy Stack

Layer 5 (Active): Large-scale Navigation (Devin Search / 1M+ LOC)

Layer 4 (Active): Vertical Workflows (Anthropic Skills / MCP)

Layer 3 (Passive): Discoverable Docs (llms.txt)

Layer 2 (Passive): Scoped Rules (.instructions.md)

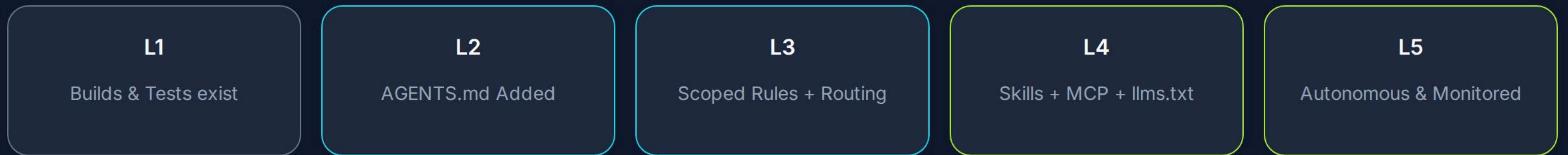
Layer 1 (Passive): Foundation (AGENTS.md / copilot-instructions.md)

Passive Steering Workflow



Result: Deterministic, framework-aligned output 100% of the time.

Repository Maturity Model



| The 15-Minute Audit

✓ **Foundation:** Create root AGENTS.md

✓ **Commands:** Add Build/Test/Lint scripts

✓ **Rules:** Add .github/copilot-instructions.md

✓ **Scope:** Set applyTo glob patterns

✓ **Routing:** Add nested AGENTS.md (monorepo)

✓ **Verify:** Ask Agent "How do I build?"

Commit. Push. You are AI-Native.

What goes in **AGENTS.md**?

Overview

1-sentence summary of the tech stack and project goal.

Commands

Execution-ready commands for setup, build, and tests.

Style

Non-obvious conventions (e.g., "Use Zod for all schemas").

Testing

How to validate changes before submitting a PR.

Pitfalls

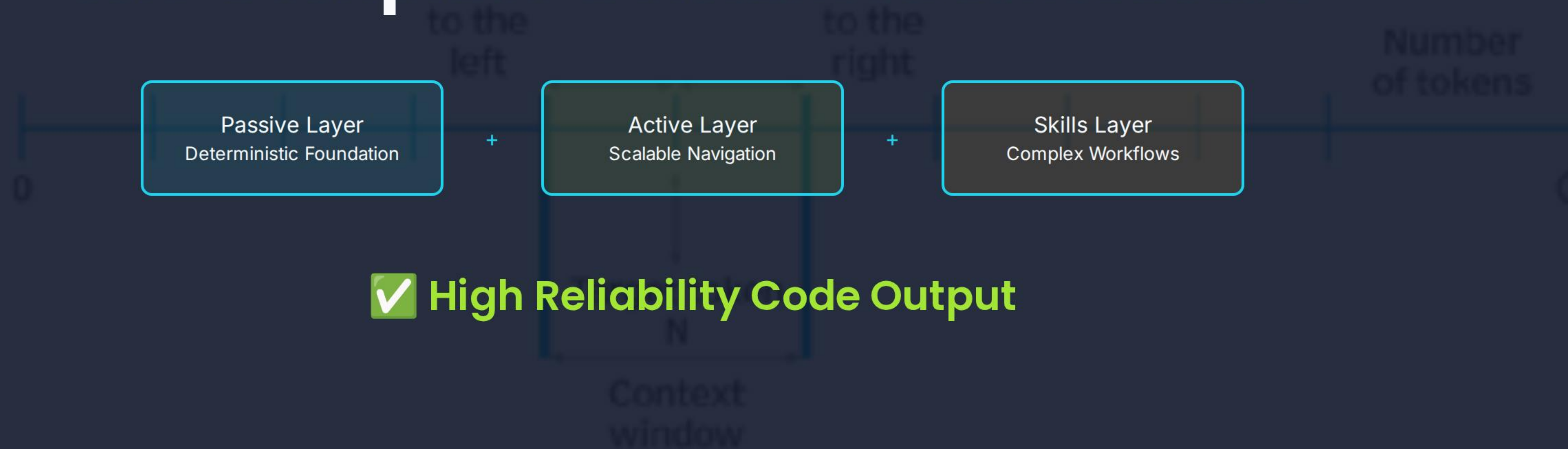
"Don't do this" list. Tribal knowledge traps.

Rule #1

Keep under 8KB (compressed).
Concise > Comprehensive.

Example of a context window

The Complete Context Architecture



Ready for AI-Agents?

Implement AGENTS.md this afternoon.

KEY SOURCES

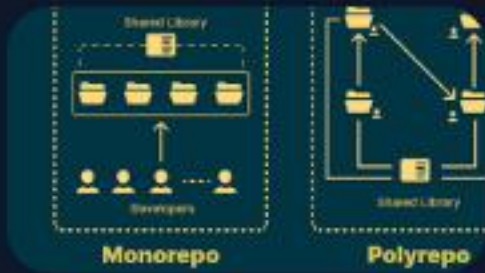
- Vercel: AGENTS.md vs Skills Eval (2026)
- arXiv 2602.11988: Evaluating AGENTS.md (ETH)
- Anthropic: Agent Skills Best Practices
- LLMStxt.org Specification

CONTACT

Engineering Tech Talk Series

github.com/ai-native-blueprints

Image Sources



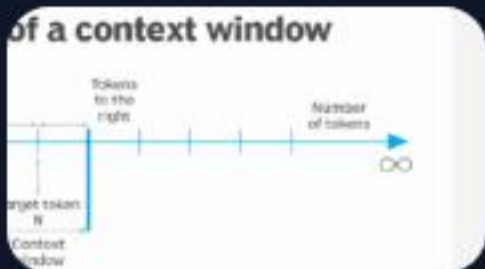
<https://www.aviator.co/blog/wp-content/uploads/2024/12/monorepo-dir.png>

Source: www.aviator.co



https://miro.medium.com/v2/resize:fit:1400/1*-qM4ZHLxi-WhrmTdTvHIfw.png

Source: [medium.com](https://miro.medium.com)



https://www.techtarget.com/rms/onlineimages/example_of_a_context_window-f_mobile.png

Source: www.techtarget.com
